

Assignment -1

1. What is NumPy? Why should we use it?

NumPy (Numerical Python) is a powerful Python library used for numerical computing. It provides support for creating and manipulating large, multi-dimensional arrays and matrices, along with a collection of mathematical functions to perform operations on them efficiently.

Why should we use NumPy?

- It performs numerical computations much faster than Python lists.
- It provides support for multi-dimensional arrays.
- It consumes less memory compared to Python lists.
- It offers built-in mathematical, statistical, and linear algebra functions.
- It is widely used in Data Science, Machine Learning, Artificial Intelligence, and Scientific Computing.
- It integrates well with libraries such as Pandas, Matplotlib, and Scikit-learn.

2. Write the steps to create 2D and 3D arrays with output.

A. Creating a 2D Array

Step 1: Import the NumPy library.

Step 2: Use `np.array()` and pass a list of lists.

Program:

```
import numpy as np

arr2d = np.array([[1, 2, 3],
                  [4, 5, 6]])
```

```
print(arr2d)
```

Output:

```
[[1 2 3]
 [4 5 6]]
```

B. Creating a 3D Array

Step 1: Import the NumPy library.

Step 2: Use `np.array()` and pass a list containing multiple 2D lists.

Program:

```
import numpy as np

arr3d = np.array([
    [[1, 2], [3, 4]],
    [[5, 6], [7, 8]]
])
```

```
print(arr3d)
```

Output:

```
[[[1 2]
  [3 4]]

 [[5 6]
  [7 8]]]
```

These examples demonstrate how to create **2D** and **3D** NumPy arrays using the `np.array()` function.